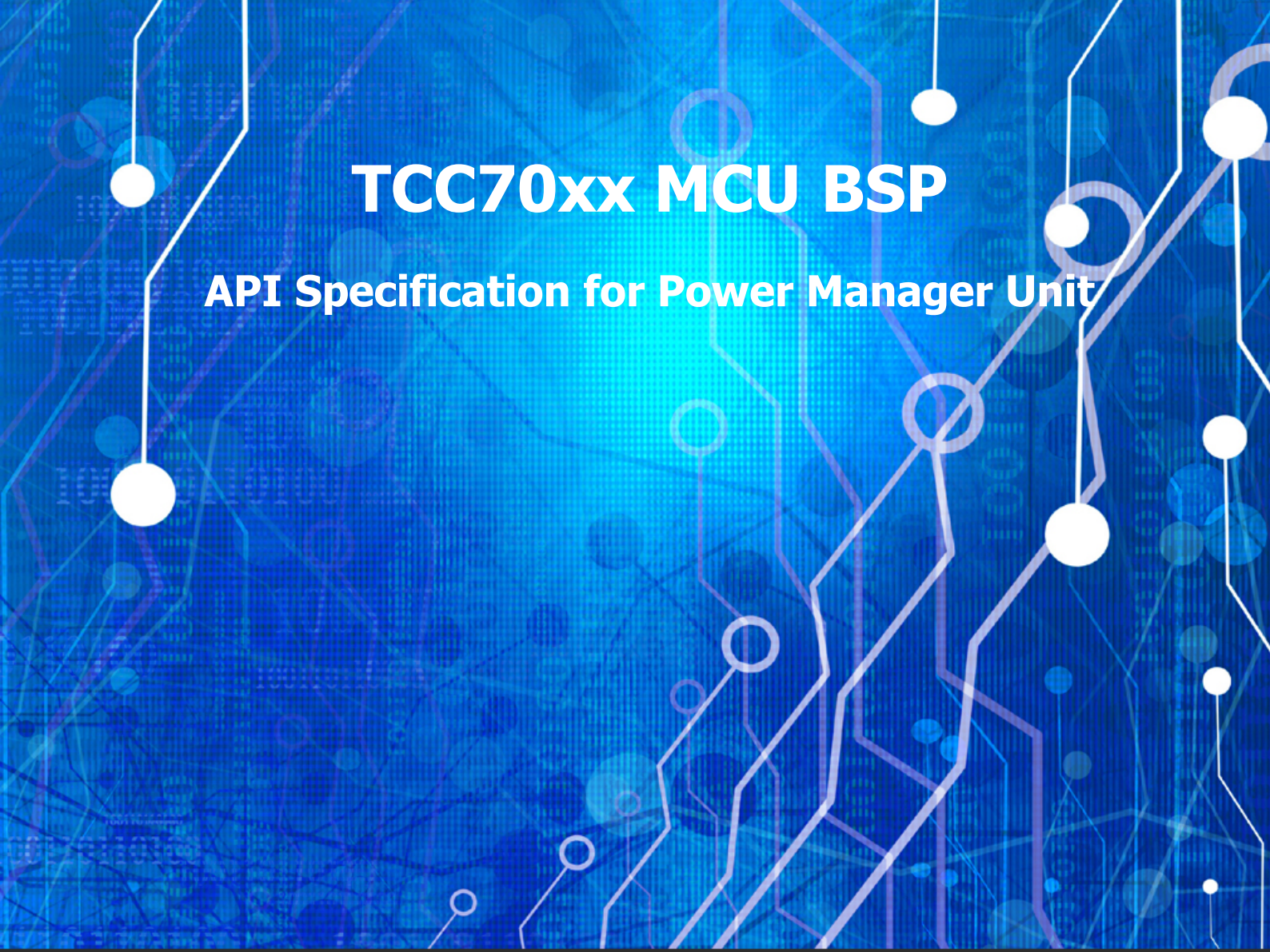




TCC70xx MCU BSP

API Specification for Power Manager Unit

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the Power Manager Unit device driver (hereinafter referred to as PMU driver) of TCC70xx MCU BSP. For more detailed information on PMU, refer to Chapter 25 Power Manager Unit (PMU) in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

PMU	Power Manager Unit
PVT	Power Voltage
FMU	Fault Management Unit

3 SOURCE FILES

3.1 Location of Source Files

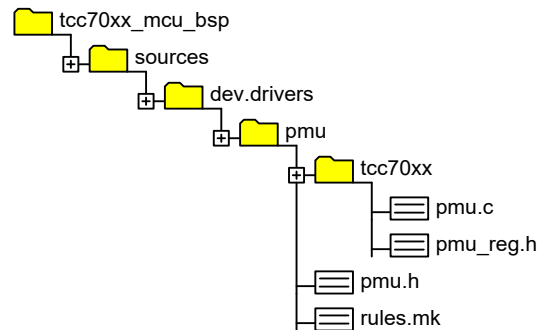


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- pmu.c
 - PMU driver source
- pmu_reg.h
 - PMU register define header
- pmu.h
 - PMU header for parameter define and interface
- rules.mk
 - Rules used for build

4 ENUMERATIONS

4.1 PMUSrc_t

Enumeration variables to manage reset source or monitoring source. This value can be used as a parameter in `PMU_GetResetOccur()`.

Declaration

```
typedef enum
{
    PMU_SRC_NOTHING    = 0,
    PMU_SRC_COLD       = 1,      //ResetOccur,
    PMU_SRC_HSM_COLD   = 2,      //ResetOccur, ResetMonitor
    PMU_SRC_FMU_FAULT  = 3,      //ResetOccur, ResetMonitor
    PMU_SRC_WARM       = 4,      //ResetOccur,
    PMU_SRC_HSM_WARM   = 5,      //ResetOccur, ResetMonitor
    PMU_SRC_WDT        = 6,      //ResetOccur, ResetMonitor
    PMU_SRC_PMU_WDT    = 7,      //ResetOccur, ResetMonitor
    PMU_SRC_HSM_WDT    = 8,      //ResetOccur, ResetMonitor
    PMU_SRC_XIN_PVT    = 10,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR0_PVT   = 11,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR1_PVT   = 12,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR2_PVT   = 13,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR3_PVT   = 14,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR4_PVT   = 15,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR5_PVT   = 16,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR6_PVT   = 17,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR7_PVT   = 18,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR8_PVT   = 19,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR9_PVT   = 20,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR10_PVT  = 21,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR11_PVT  = 22,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR12_PVT  = 23,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR13_PVT  = 24,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_PWR14_PVT  = 25,     //ResetOccur, ResetMonitor, ClearPvtError
    PMU_SRC_MAX        = 26,
} PMUSrc_t;
```

Description

Value	Description
<code>PMU_SRC_[name]</code>	Value that defines PMU sources

Remark

None

4.2 PMUMonSrc_t

Enumeration variables to manage monitoring source. This value can be used as a parameter in PMU_EnableResetMonitor(), PMU_DisableResetMonitor().

Declaration

```
typedef enum
{
    PMU_MON_SRC_NOTHING      = PMU_SRC_NOTHING ,
    PMU_MON_SRC_HSM_COLD     = PMU_SRC_HSM_COLD ,
    PMU_MON_SRC_FMU_FAULT    = PMU_SRC_FMU_FAULT,
    PMU_MON_SRC_HSM_WARM     = PMU_SRC_HSM_WARM,
    PMU_MON_SRC_WDT          = PMU_SRC_WDT,
    PMU_MON_SRC_PMU_WDT      = PMU_SRC_PMU_WDT,
    PMU_MON_SRC_HSM_WDT      = PMU_SRC_HSM_WDT,
    PMU_MON_SRC_XIN_PVT      = PMU_SRC_XIN_PVT,
    PMU_MON_SRC_PWR0_PVT     = PMU_SRC_PWR0_PVT ,
    PMU_MON_SRC_PWR1_PVT     = PMU_SRC_PWR1_PVT ,
    PMU_MON_SRC_PWR2_PVT     = PMU_SRC_PWR2_PVT ,
    PMU_MON_SRC_PWR3_PVT     = PMU_SRC_PWR3_PVT ,
    PMU_MON_SRC_PWR4_PVT     = PMU_SRC_PWR4_PVT ,
    PMU_MON_SRC_PWR5_PVT     = PMU_SRC_PWR5_PVT ,
    PMU_MON_SRC_PWR6_PVT     = PMU_SRC_PWR6_PVT ,
    PMU_MON_SRC_PWR7_PVT     = PMU_SRC_PWR7_PVT ,
    PMU_MON_SRC_PWR8_PVT     = PMU_SRC_PWR8_PVT ,
    PMU_MON_SRC_PWR9_PVT     = PMU_SRC_PWR9_PVT ,
    PMU_MON_SRC_PWR10_PVT    = PMU_SRC_PWR10_PVT,
    PMU_MON_SRC_PWR11_PVT    = PMU_SRC_PWR11_PVT,
    PMU_MON_SRC_PWR12_PVT    = PMU_SRC_PWR12_PVT,
    PMU_MON_SRC_PWR13_PVT    = PMU_SRC_PWR13_PVT,
    PMU_MON_SRC_PWR14_PVT    = PMU_SRC_PWR14_PVT,
    PMU_MON_SRC_MAX          = PMU_SRC_MAX
} PMUMonSrc_t;
```

Description

Value	Description
PMU_MON_SRC_[name]	Value that defines PMU monitoring sources

Remark

None

4.3 PMUPvtSrc_t

Enumeration variables to manage PVT source. This value can be used as a parameter in PMU_ClearPvtError().

Declaration

```
typedef enum
{
    PMU_PVT_SRC_NOTHING    = PMU_SRC_NOTHING,
    PMU_PVT_SRC_XIN_PVT    = PMU_SRC_XIN_PVT,
    PMU_PVT_SRC_PWR0_PVT   = PMU_SRC_PWR0_PVT,
    PMU_PVT_SRC_PWR1_PVT   = PMU_SRC_PWR1_PVT,
    PMU_PVT_SRC_PWR2_PVT   = PMU_SRC_PWR2_PVT,
    PMU_PVT_SRC_PWR3_PVT   = PMU_SRC_PWR3_PVT,
    PMU_PVT_SRC_PWR4_PVT   = PMU_SRC_PWR4_PVT,
    PMU_PVT_SRC_PWR5_PVT   = PMU_SRC_PWR5_PVT,
    PMU_PVT_SRC_PWR6_PVT   = PMU_SRC_PWR6_PVT,
    PMU_PVT_SRC_PWR7_PVT   = PMU_SRC_PWR7_PVT,
    PMU_PVT_SRC_PWR8_PVT   = PMU_SRC_PWR8_PVT,
    PMU_PVT_SRC_PWR9_PVT   = PMU_SRC_PWR9_PVT,
    PMU_PVT_SRC_PWR10_PVT  = PMU_SRC_PWR10_PVT,
    PMU_PVT_SRC_PWR11_PVT  = PMU_SRC_PWR11_PVT,
    PMU_PVT_SRC_PWR12_PVT  = PMU_SRC_PWR12_PVT,
    PMU_PVT_SRC_PWR13_PVT  = PMU_SRC_PWR13_PVT,
    PMU_PVT_SRC_PWR14_PVT  = PMU_SRC_PWR14_PVT,
    PMU_PVT_SRC_MAX        = PMU_SRC_MAX
} PMUPvtSrc_t;
```

Description

Value	Description
PMU_PVT_SRC_[name]	Value that defines PMU PVT sources

Remark

None

5 APIs

5.1 PMU_Init

This function is called from *bsp.c*, and PMU is initialized in BSP initialization.

Declaration

```
void PMU_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.2 PMU_GetResetOccur

Gets the reset status for the reset source.

Declaration

```
boolean PMU_GetResetOccur
(
    PMUSrc_t   tRst
);
```

Parameters

Value	Description
tRst	[in] Source to check reset

Return

Value	Description
boolean	false: non-occurrence true: occurrence

Remark

None

5.3 PMU_EnableResetMonitor

Enables reset monitoring.

Declaration

```
void PMU_EnableResetMonitor
(
    PMUMonSrc_t    tRst
);
```

Parameters

Value	Description
tRst	[in] Monitoring source for reset or FMU interrupt

Return

None

Remark

None

5.4 PMU_DisableResetMonitor

Disables reset monitoring.

Declaration

```
void PMU_DisableResetMonitor
(
    PMUMonSrc_t    tRst
);
```

Parameters

Value	Description
tRst	[in] Monitoring source for reset or FMU interrupt

Return

None

Remark

None

5.5 PMU_ColdReset

Performs a cold reset.

Declaration

```
void PMU_ColdReset  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.6 PMU_WarmReset

Performs a warm reset.

Declaration

```
void PMU_WarmReset  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.7 PMU_ClearPvtError

Clears the PVT error state.

Declaration

```
void PMU_ClearPvtError
(
    PMUPvtSrc_t    tPvt
);
```

Parameters

Value	Description
tPvt	[in] PVT source to clear error status

Return

None

Remark

None

6 HOW TO USE PMU DRIVER

6.1 Initialize PMU Driver

```
{  
    PMU_Init();  
}
```

6.2 Get Reset Status

```
mcu_printf(" - PMU_RESET_COLD      = %s.\n", PMU_GetResetOccur(PMU_SRC_COLD    ) == 1 ? "1 !" : "0");
```

6.3 Set Reset Monitoring

```
PMU_EnableResetMonitor(PMU_SRC_XIN_PVT);  
PMU_DisableResetMonitor(PMU_MON_SRC_XIN_PVT);
```

6.4 Set Cold Reset

```
PMU_ColdReset();
```

6.5 Set Warm Reset

```
PMU_WarmReset();
```

6.6 Clear PVT Reset Status

```
PMU_ClearPvtError(PMU_SRC_XIN_PVT);
```

7 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

8 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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